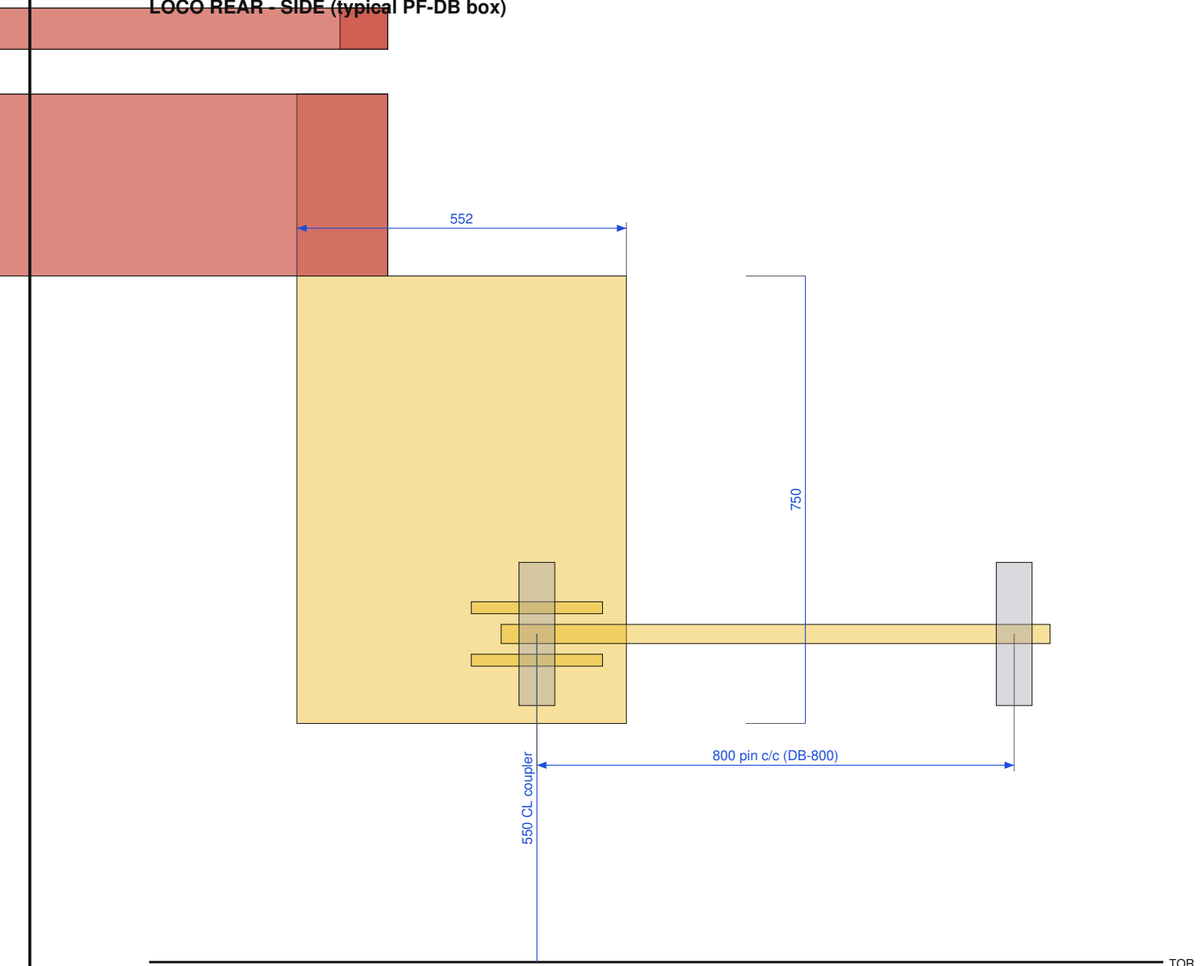
LOCO REAR - SIDE (typical PF-DB box)



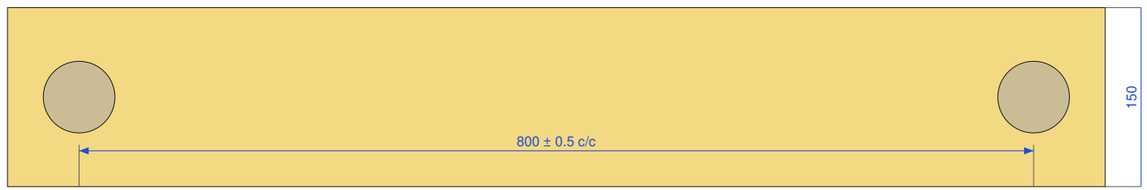
DRAWGEAR SCHEDULE (one design, used on every vehicle)

Item	Spec	Size	Note
DB-800	PL32x150 A572-50	800 c/c	Loco-tender
DB-1473	PL32x150 A572-50	1,473 c/c	Tender-coach, coach-coach
Pin	Ø60 4140 Q&T, H8/f7	L=150 u/head	grip 108 + head 20 + Ø10 cross-bolt + R-clip
Bushing	Ø61/Ø75 spacer, 4140	2 per pin	limits gap to 4 mm
Chains	2× G100 5/8 in, WLL 22,600 lbf ea.	slack 150	2× G-2130 1-1/8 + 2× S-1325A (L4.2); loco lugs PL44.5
Box	PL16 A572-50	400 wide	10 mm fillet all round + MT

WHY / CHECKS

Loads (owner basis, PE to confirm): ±35 kN normal; 70 kN no permanent deformation; 200 kN ultimate (no rupture).
Checks: drawbar net 70 MPa, bearing 104 MPa, buckling 2.7×; pin bending 104 MPa, shear 35 MPa; plates bearing 83 MPa; box 34 MPa, weld 85 MPa (all at 200 kN). Chains MBS 804 kN.
Why: replaces the 752 mm eccentric rear mount and the 400 mm tender offset with one closed box that carries the moment straight into both rails via knee gussets.
Truck restraint (I-08) stays with the PE: centre-pin retaining collar + 2 safety straps per truck shown on L-103/C-303.

DB-800 PLAN 1:5



PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

PF-DB drop drawbar box, drawbar, pin and safety chains

Sheet L-104 · REV L4.2 · 2026-10-28 · 1:10 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED